

MULTI-STAGE AI PROMPT WORKFLOW

Patient Intake Triage Priority Prediction and Escalation

1. Selected Scenario

This assessment uses a patient-intake triage workflow in which AI converts free-form patient presentations into an organized priority assessment, prepares an operational response, and then reviews that response through a further AI critique stage.

The workflow also tracks whether a patient has returned for the same complaint and how long the patient has been waiting. A repeat presentation is treated as an additional urgency indicator because it may reflect a worsening or unresolved problem. When the permitted waiting period is exceeded without assessment, the case is escalated to the attending physician.

Training note: This is a prompt-chain design exercise for a patient intake queue. It is not intended to function as clinical decision support and does not substitute for evaluation by a qualified clinician.

2. Objectives

- Assign appropriate triage categories to incoming patient presentations.
- Recognize repeated visits and include them as an urgency factor.
- Produce an actionable plan for the intake and triage staff.
- Identify cases requiring physician escalation after the permitted waiting period.
- Demonstrate a three-stage chained prompting process containing a mandatory critique step.

3. Patient Input Dataset

Patient	Presentation	Location	Repeat visit	Waiting time
P001	Crushing chest pain extending to the left arm, with sweating and shortness of breath.	Room A	No	10 min
P002	Mild seasonal allergy symptoms including sneezing and itchy eyes.	Room B	No	45 min
P003	Child with a 104°F fever who is lethargic and difficult to wake; second visit today.	Room C	Yes	15 min
P004	Ankle sprain following a fall, with swelling; patient can bear some weight.	Room D	No	40 min
P005	Abdominal pain that is getting worse; reported during three visits this week.	Room E	Yes	30 min

4. Prompt Chain and AI Outputs

Step 1 — Information Extraction and Initial Planning

Prompt 1: Review the supplied patient intake cases as an emergency-department triage analyst. Extract the factors that should influence priority, including possible life-threatening conditions, vulnerable patients such as children, repeated visits, symptom severity, and waiting duration. Build a clear triage framework and apply it to every case. Indicate which cases require physician escalation if they are not assessed within the allowed period. Return a table containing Patient ID, Triage Level, Reason, Suggested Action, and Escalation Condition. Cases: P001 · chest pain radiating to the left arm with sweating; P002 · mild seasonal allergies; P003 · child with 104°F fever and lethargy, second visit today; P004 · swollen sprained ankle after a fall; P005 · worsening abdominal pain across three visits this week.

Step 1 AI Result:

ID	Level	Reason	Action	Escalation
P001	Critical	Chest pain with radiation, sweating and breathing difficulty indicates a potentially serious cardiac pattern.	Transfer to the resuscitation area immediately, check ECG and vital signs, and notify the physician.	Immediate physician escalation; no waiting period.
P003	Critical	A very high fever with lethargy in a child is a warning sign, made more significant by the repeat visit.	Arrange prompt physician assessment and reassess vital signs and hydration.	Escalate if assessment has not occurred within 15 minutes.
P005	High	Worsening abdominal pain together with repeated visits may indicate an unresolved or deteriorating condition.	Perform a complete abdominal assessment, consider investigations, and review earlier visit records.	Escalate if still unresolved or unassessed after 30 minutes.
P004	Medium	The injury affects the ankle but the patient retains some ability to bear weight and there is no stated immediate life or limb threat.	Consider X-ray when appropriate, along with splinting and pain management.	Escalate if unresolved after 90 minutes.
P002	Low	The symptoms are mild and seasonal with no stated immediate safety concern.	Provide symptomatic management or antihistamine guidance.	Escalate if symptoms worsen or waiting exceeds 3 hours.

Step 2 — Response Plan Generation

Prompt 2: Based only on the framework and patient assessment from Step 1, prepare a practical intake response plan for the care team. Keep the same triage categories. Explain the order in which patients are handled, how repeat visits raise urgency, and how an unassessed patient is escalated after the relevant time limit. Add a concise physician alert that can be issued when escalation is triggered. The plan should be appropriate for a hospital intake/triage workflow.

Step 2 AI Result: The queue is handled in the order Critical, High, Medium, then Low. Presentations involving an immediate threat to life receive the greatest priority. A patient who repeatedly returns with the same or worsening complaint receives additional attention because repeated visits can signal deterioration. Vulnerable patients, including children with high fever, receive urgency comparable to adult life-threat presentations.

The system records each patient's intake time, current status, triage category, number of related previous visits, assigned clinician, and assessment time. It continually compares unassessed cases with their escalation limits. Once a limit is exceeded, the attending physician receives an automatic notification.

Example physician alert: ALERT: Patient P003 has passed the 15-minute assessment limit. Presentation: pediatric high fever with lethargy. Related visits: 2. Triage: CRITICAL. Immediate assessment requested.

Step 3 — Critique and Refinement Gate

Prompt 3: Act as a senior reviewer before the response plan is finalized. Check the Step 2 draft for five points: whether life-threatening and vulnerable-patient cases are clearly prioritized; whether repeated visits are explicitly used as an urgency signal; whether the response explains the action taken after an assessment deadline is missed; whether the physician alert contains enough information to act; and whether the workflow is practical for hospital intake. First identify weaknesses, then provide a corrected version without changing the core information from Step 1.

Step 3 Review: The draft appropriately gives priority to life-threatening and vulnerable-patient cases. The repeat-visit rule should be stated more directly. The escalation mechanism should explicitly say that unassessed cases are checked continuously against their waiting limits. The physician alert is useful but should clearly include the triage level, number of repeat visits, and elapsed waiting time.

Refined Final Workflow

The intake process first evaluates life-threat risk, patient vulnerability, symptom severity, repeat presentations, and the remaining time before escalation. Immediate life threats and high-risk signs in vulnerable patients are categorized as Critical. High-severity cases are categorized as High when they present a safety concern or repeated pattern. Medium and Low categories cover presentations with lower immediate risk.

Repeated visits add urgency to the assessment. When a patient returns several times for the same or worsening complaint, the repeated pattern is flagged so that it receives greater attention than an isolated presentation of similar general severity.

Once a patient is assigned to a care area or clinician, the system continues monitoring whether an assessment has taken place. If the configured waiting limit is crossed, the attending physician is automatically alerted.

Final notification example: ALERT · Unassessed Patient. Patient ID: P003. Presentation: pediatric high fever with lethargy. Triage Level: CRITICAL. Previous related visits: 2. Waiting time: 15+ minutes. Status: Unassessed. Required action: Please assess and assign immediate care.

5. Workflow Summary

Stage	Input	System / AI activity	Output
1. Extraction / Planning	Unstructured patient intake information	Extract relevant triage factors and determine priority.	Triage framework and priority table
2. Generation / Drafting	Results from Stage 1	Convert the analysis into a practical care-team plan and escalation process.	Operational plan and physician notification
3. Evaluation / Refinement	Draft from Stage 2	Check the draft against the review criteria and improve it.	Refined final workflow

6. Expected System Results

- Patients displayed according to triage urgency.
- Critical, High, Medium, or Low priority labels.
- Recognition of repeated visits involving the same or related complaint.
- Allocation of patients to the appropriate clinician or care area.
- Monitoring of patients who have not yet been assessed and their elapsed wait.
- Automatic physician notification when an escalation threshold is exceeded.
- A staff dashboard showing the most urgent and overdue cases first.

7. Assessment Requirement Check

Requirement	Result
Minimum of three chained prompts	Met — three sequential prompts are provided.
Dependency between prompts	Met — Step 2 is based on Step 1 and Step 3 evaluates/refines Step 2.
Critique gate	Met — Step 3 uses a defined review checklist.
Exact prompts documented	Met — all three prompts are included.
Intermediate AI responses	Met — outputs from each stage are recorded.
Live chat evidence	A screenshot from the actual AI interface must be inserted in Section 7.

8. Conclusion

This example shows how a patient-intake triage task can be organized through a sequence of AI prompts instead of one standalone request. Stage 1 structures the intake information, Stage 2 turns that analysis into an operational plan, and Stage 3 checks and improves the plan. The workflow also incorporates repeat visits and delayed assessment as signals that can lead to increased urgency and physician escalation.